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Team name	A9
Project Title	Lumos

Lumos Project

Smart Wearables Prototyping

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Demo Link: https://youtu.be/2R6Sbul6aNU?is=nBhr0ABMkU_S1Qfo

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I. Individual Contribution

Lorenzo Maccarini: My contributions mainly concerned the hardware design of the Lumos sensor board and the firmware development for the microphone and light sensor. I worked on schematic design and PCB design, and implemented the firmware routines used for sensor acquisition, calibration, and validation.

Lara Dau: My contributions to the project began with the initial brainstorming sessions and went on to define the core tracking aims, with a focus on optimal sensor integration and background research for the thresholds regarding focus and stress values to the different sensors. One of the technical challenges I addressed was trying to configure the digital microphone communication to capture and process acoustic data. I also tried to develop a calibration strategy for the light sensor to estimate ambient colour temperature. I concluded my part by documenting in the technical report and creating the demonstration video.

Luca Tufano: my contributions regarded the discussion of the main objectives of the board, I participated in the debugging phase of the first firmware, and my main contribution was the development of the step counter and the color temperature identification algorithm. I also helped by synthesizing some sections of the report and writing some of them myself.

Davide Venturini: I mainly focused on the development of the software, Playing an important role in the implementation of the step counter and the measurement of the light temperature both in the modeling phase and in the integration within the firmware. I also reworked the processing of the data of the IMU, handled and participated in several testing and debugging sessions. Other contributions include report sections and the planning and production of the video demo.

Youssef Gad: I was involved in the project design, and implementation planning. I contributed in designing the sensor board on Altium (schematics and PCB), then in programming the firmware of the mainboard (acquisition of sensor data, processing data metrics, and testing and validation). Finally, I was involved in all stages of designing and implementing the flutter app beside the documentation for the report.

II. Project Aims

*Lara Dau (100%), Luca Tufano (100%), Davide Venturini (100%), Youssef Gad (100%),
Lorenzo Maccarini (100%)*

The primary objective of this project is to develop a smart glasses prototype that monitors environmental and behavioral factors to optimize user concentration. Equipped with a microphone, a light sensor, and an IMU, the wearable device tracks acoustic levels, light temperature, light exposure, and physical activity, also transmitting this data via Bluetooth Low Energy to a mobile application for real-time cognitive analysis.

Specifically, the project aims to classify gathered data into critical focus and stress levels:

- **Acoustic and Visual Monitoring:** The device evaluates noise levels and light parameters to detect stress-inducing deviations. It identifies optimal ranges for concentration, while monitoring how excessive ambient noises, light temperatures and light exposure interfere with long-term stress regulation and alertness.
- **Physical Activity Tracking:** Utilizing the IMU, the device tracks steps to help the user achieve the ideal step range necessary for optimal cognitive regeneration during study breaks or to keep the brain awake while studying.

Ultimately, the project aims to empower users to actively adapt their environments and behavioral habits for sustained cognitive performance.

III. Hardware Description

Lara Dau (20%), Luca Tufano (25%), Davide Venturini (40%), Youssef Gad (100%), Lorenzo Maccarini (100%)

a. Core Architecture and Component Selection

The flex bridge connector: flexible board that enables communication between the main board and the external probes (in both I²C and PDM protocol).

The light sensor (AS7341): provide spectral identification of the entering light wave; it presents 8 channels to measure visible light and 3 other channels to measure NIR light, frequency of the entering light (Flicker) and general entering light intensity (Clear); the data is transmitted with I²C protocol.

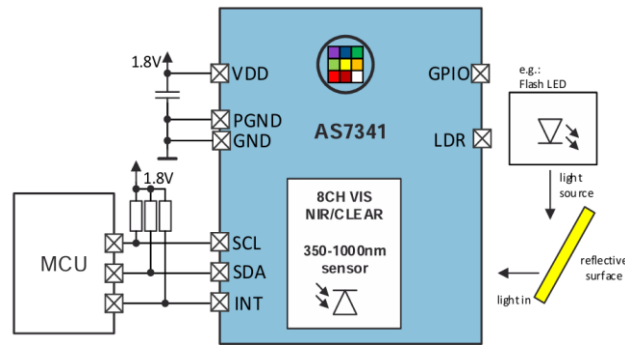


Figure 1, AS7341 sensor diagram showing pin configurations

The microphone (IMP34DT05): ultra-compact, low-power, omnidirectional, digital MEMS that detect air vibrations through a silicon micromachining process and then transmit the data in PDM format

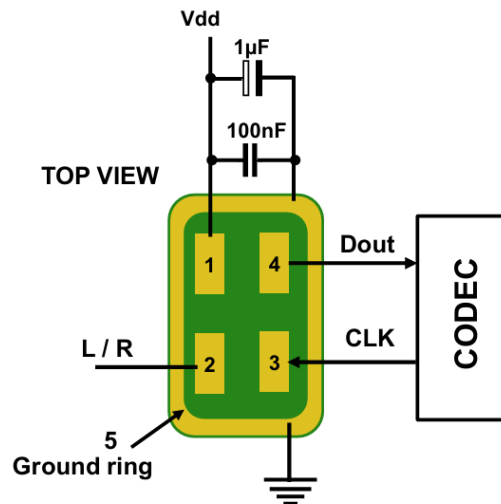


Figure 2, IMP34DT05 Microphone diagram and pin configuration

b. Circuit Design

In the design of the LUMOS probe, the main objective was to obtain a compact circuit suitable for integration in a smart eyewear application.

Since wearable devices are strongly constrained by size and power consumption, the circuit was kept as simple as possible while preserving the required sensing and communication functionalities.

It is important in this process to maintain consistency inside the connection between the Narwhal board, the flex bridge and the LUMOS probe; here we can see the final schematics that we have built:

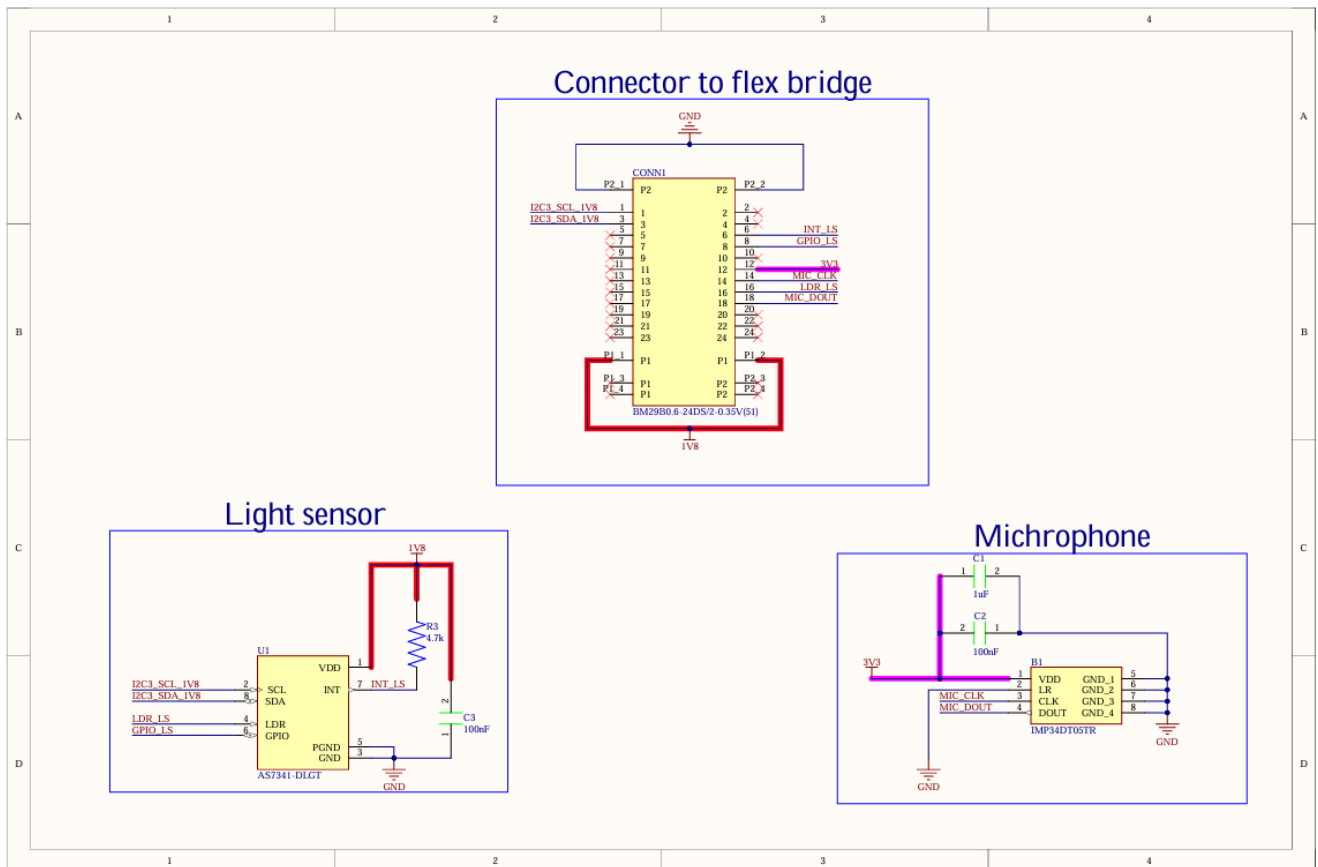


Figure 3, Lumos Board schematic designed on Altium Designer

The AS7341 light sensor is supplied from the 1.8V rail and communicates with the MCU through the I²C bus. The schematic also routes the INT_LS, GPIO_LS, and LDR_LS lines, which can be used for interrupting generation, synchronization, or light-source control. It also includes a 100nF decoupling capacitor between VDD and ground to stabilize the 1.8V supply and reduce high-frequency noise and a 4.7k Ω resistor connected to the INT_LS line as a pull-up resistor for the open-drain interrupted output of the sensor (both recommendations of the light datasheet).

The IMP34DT05 is connected as a digital MEMS microphone: the MIC_CLK signal provides the synchronization clock, MIC_DOUT carries the PDM output stream, and MIC_LR selects the valid clock phase/channel configuration. The microphone datasheet defines V_{dd}, LR, CLK, DOUT, and GND as the main pins and specifies a digital PDM output; in addition, two decoupling capacitors are included on the supply line, with values of 100nF and 1 μ F, recommended by the manufacturer.

c. PCB Layout Design

Based on the guidelines provided, we designed the sensor board using Altium Designer. This board is connected to the mainboard using a flexible bridge connector (BM29B0.6-24DS/2-0.35V), see Figure 4.

Dimensions: The sensor board is extremely compact, measuring just 11.05 mm (L) x 8.26 mm (W) x 1.01 mm (H). We chose a 1.01 mm PCB thickness because it perfectly matches the minimum limits allowed by our manufacturer, JLCPCB.

Layer Stackup: We used a standard 2-layer design. The Top Layer holds all the sensors, resistors, and capacitors. The Bottom Layer holds the connector for the flex bridge. The two layers are connected using standard, through hole vias.

Routing & Manufacturing: We configured our Altium design rules to strictly follow JLCPCB's manufacturing limits. To save time and ensure correct connections, we utilized Altium's auto-routing tool across both layers.

Grounding: To help reduce electrical noise and provide a solid reference for the sensors, we added copper polygon pours assigned to the GND net on both the top and bottom layers and connected through a normal through hole via.

We have also taken other key design points into account such as, connector oriented to be in a separate layer, and for the sensors to be on the opposite layer to face outwards to the environment and keeping short and direct routing for I²C and PDM pins.

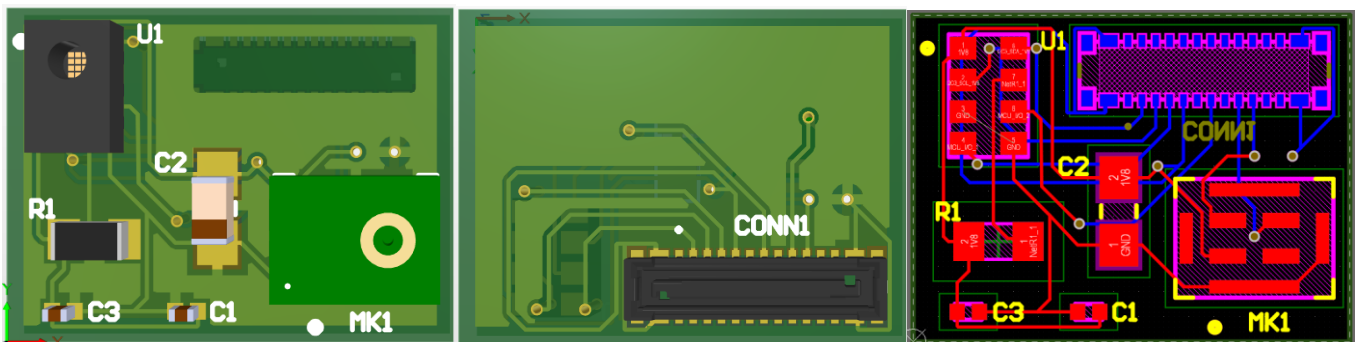


Figure 4, Front/Back View of the PCB designed on Altium Designer

d. Production Costs

The prototype was assembled using off-the-shelf development modules for rapid iteration. The estimated unit cost for a short-run production is as follows:

Table 1, Estimated prototype BOM cost for short-run production

Component / Module	Function in Prototype	Estimated Cost
STM32 Microcontroller	Main processing unit	€ 8,55
BQ25185 Battery Charger	Battery charging and power management	€ 1,80
6-Axis IMU	Motion sensing unit	€ 4,92
RN4871 BLE Module	Bluetooth Low Energy communication	€ 7,95
Flash Memory	Non-volatile data storage	€ 28,53
AS7341 Spectral Sensor	Multi-channel optical sensing	€ 6,77
IMP34DT05TR Microphone	Digital MEMS microphone	€ 2,07
BM29B0.6-24DS/2-0.35V(51)	Connector / interconnect component	€ 1,21
Passive Components & PCB Fab.	Passives and PCB fabrication	€ 2,08
Total Estimated BOM Cost		€ 63,88

Note. Costs represent the prototype phase and are expected to scale down significantly in mass manufacturing.

Charge Details		^
Special Offer		€1.74
Via Covering		€0.00
Surface Finish		€0.00
Deburring/Edge rounding		€0.17
PCB Build Time ?		
☒ 2 days		€0.00
☐ 24 hours		€6.54
☐ 24 hours	PCBA Only	€0.00
Calculated Price		€1.92
Additional charges may apply for special cases		

Figure 5, Screenshot of JLCPCB website showing the quote for the Lumos PCB

IV. Firmware Design

Lara Dau (30%), Luca Tufano (80%), Davide Venturini (80%), Youssef Gad (80%), Lorenzo Maccarini (80%)

The firmware developed for the smart glasses prototype is implemented on the microcontroller (STM32U575RIT6Q) to ensure tracking in terms of physical activity,

concentration and stress factors. The firmware architecture is divided into four stages: peripheral configuration, asynchronous data acquisition, real-time on-board data processing and BLE data transmission to the smartphone app.

a. Data Acquisition

The main loop is based on a two-phase acquisition in a 2.5s cycle:

- 500ms for environmental sensing, including light and audio;
- 2000ms for IMU data acquisition;

This way, the sensors don't block each other, and we can still send Bluetooth updates smoothly at 2 Hz.

Here is how the main loop is structured to handle the sensors:

Light Sensor (AS7341)

The light sensor reads 11 different colors of light, but it only has 6 ADCs. To get all the colors, we must reconfigure its internal multiplexer and read it in two separate steps over I²C. Since waiting for the sensor to finish takes over 50ms, we built a non-blocking state machine. This lets the CPU start the measurement and walk away to do other tasks instead of waiting.

```
// --- Continuously Drive Light Sensor ---  
LightMetrics_ProcessPendingSample();  
if (light_was_pending && LightMetrics_HasPendingSample() == 0)  
{  
    // Request the next sample immediately to keep accumulating  
    LightMetrics_RequestSample();  
    light_was_pending = 1;  
}
```

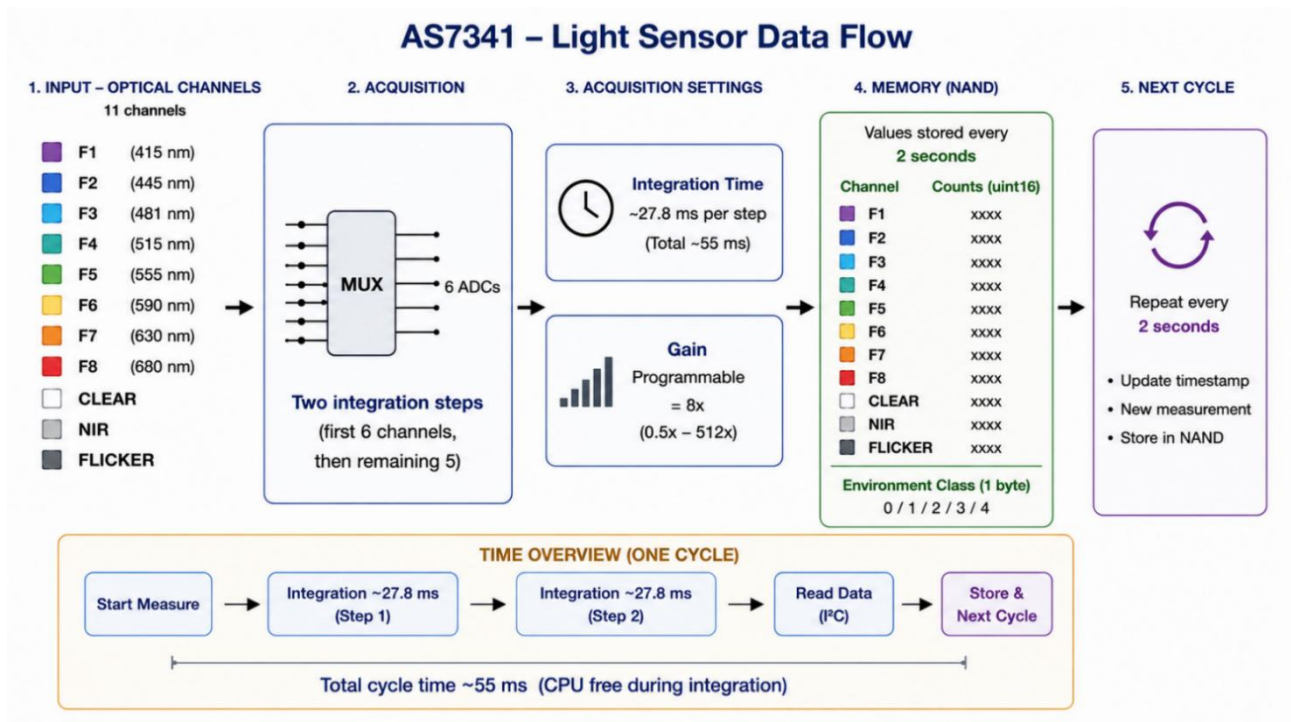


Figure 6, AS7341 light sensor data flow

Microphone (IMP34DT05TR)

We used STM32's Multi-Function Digital Filter (MDF) to read the microphone combined with Direct Memory Access (DMA), which puts the audio directly into the RAM to keep the CPU in sleep state and saves power.

During the 500ms phase, the DMA tells the CPU when the buffer is full. The CPU then copies the audio into a ring buffer, and the main loop drains it to calculate the noise level in decibels (dBA) using an A-Weighting filter.

```

while (AudioRing_Count(&g_audio_ring_buffer) >= AUDIO_CHUNK_SAMPLES)
{
    static int16_t chunk[AUDIO_CHUNK_SAMPLES];
    uint32_t popped = AudioRing_PopChunk(&g_audio_ring_buffer, chunk,
        AUDIO_CHUNK_SAMPLES);
    if (popped > 0) {
        uint16_t calc_laeq;
        uint8_t calc_env;
        MicMetrics_ProcessFrame(chunk, popped, &calc_laeq, &calc_env);
        laeq_sum += calc_laeq;
        laeq_count++;
        g_last_audio_env = calc_env; } }
  
```

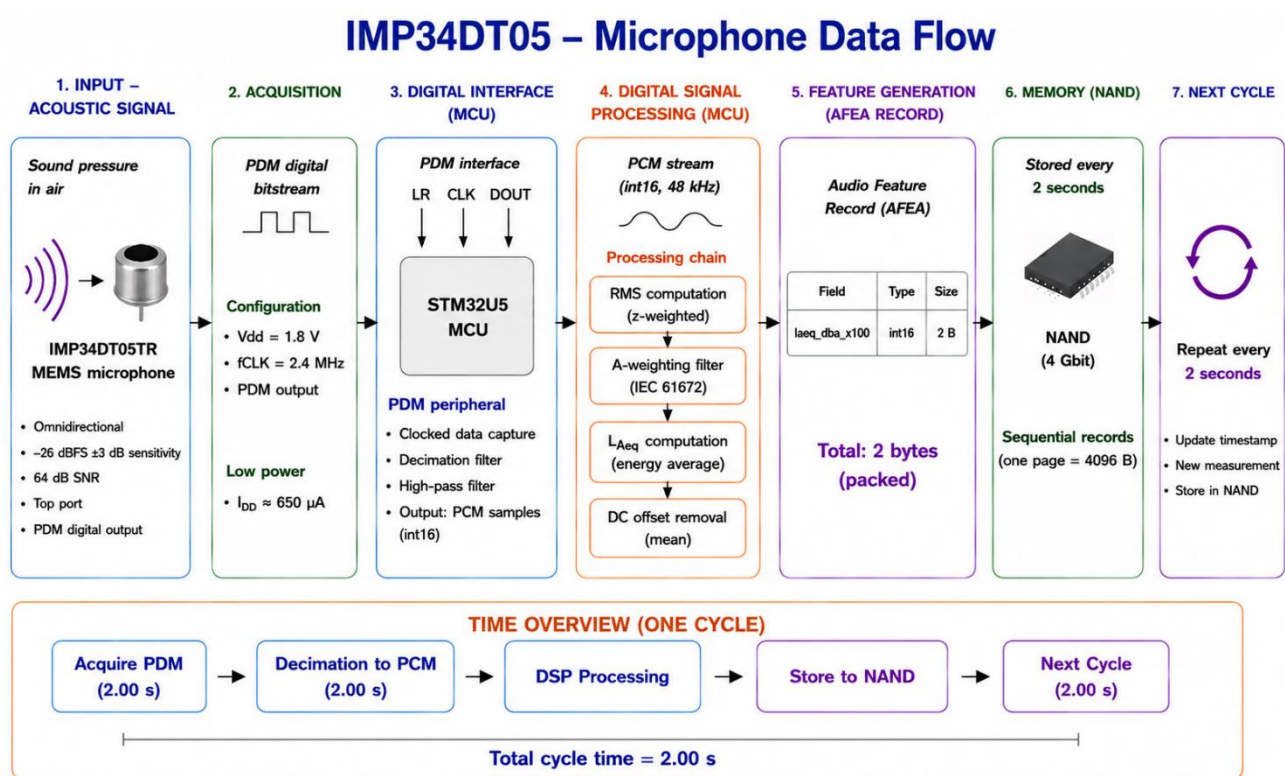


Figure 7, Complete audio acquisition and processing chain

Inertial Sensor (LSM6DSO16IS IMU)

The IMU sends acceleration data over I²C at 104 Hz. The I²C driver saves the incoming data into a circular ring buffer first. Later, in the main loop, the step counter code pulls the data out of the buffer and feeds it to the StepCounter function to update the step count.

```
while (IMU_RingBuffer_Pop(&g_imu_ring_buffer, &popped))
{
    IMU_ConvertAccelRawToFloat(&accelerometer_data, popped.acc);
    IMU_ConvertGyroRawToFloat(&gyroscope_data, popped.gyro);

    StepCounter_U.In1[0] = gyroscope_data.x;
    StepCounter_U.In1[1] = gyroscope_data.y;
    StepCounter_U.In1[2] = gyroscope_data.z;
    StepCounter_U.In2[0] = accelerometer_data.x;
    StepCounter_U.In2[1] = accelerometer_data.y;
    StepCounter_U.In2[2] = accelerometer_data.z;
}
```

Data Fusion and Bluetooth (BLE)

At the end of the cycle, we take all the processed results (steps, light class, blue light ratio, light temperature, noise dB, noise class) and pack them into a 20-byte data structure.

```
if (current_tick - imu_ble_timer >= 500)
{
    imu_ble_timer = current_tick;

    BLE_UnifiedPayload ble_payload;
    ble_payload.stepCount      = StepCounter_Y.stepnumber;
    ble_payload.light_level_class = g_last_light_class;
    ble_payload.blue_clear_ratio = g_last_blue_clear_ratio;
    ble_payload.color_temp      = g_last_color_temp;
    ble_payload.laeq_x10        = g_last_laeq_x10;
    ble_payload.audio_env_class = g_last_audio_env;

    BLE_SendUnifiedPacket(&ble_payload);
}

// --- Check for Phase Transition (2000ms) ---
if (current_tick - phase_start_ms >= 2000)
{
    current_phase = PHASE_ENV;
    phase_start_ms = current_tick;
    phase_just_entered = 1;
}
```

This payload is sent over UART to the Bluetooth module exactly 2 times per second, giving the smartphone app a small sized and structured update of the user's environment.

b. Data Processing

The core intelligence of the firmware lies in its real-time data processing and threshold mapping. Instead of streaming heavy raw datasets, the firmware evaluates the data on the microcontroller against critical cognitive zones.

Acoustic Filter: Sound levels are computed over an algorithmic window using an A-weighting filter, which approximates the frequency sensitivity of human hearing. To convert

the digital RMS value into an estimated acoustic level, a single-point calibration was performed using a 1 kHz tone generated with a Python script. From that we get:

$$L_{A,eq} = L_{A,dBFS} + 122.40$$

where $L_{A,dBFS}^{9,10}$ is the A-weighted RMS level in dBFS, and 122.40 dB is the experimental calibration offset. Although this is a single-frequency calibration, the microphone datasheet shows an approximately flat response around 1 kHz, see Figure 8, and up to about 10 kHz, making it suitable for an estimated A-weighted sound exposure measurement.

The firmware maps 45dB - 70dB as the optimal concentration range¹; readings falling below 40dB or exceeding 70dB are immediately flagged as chronic stress factor.^{2,3}

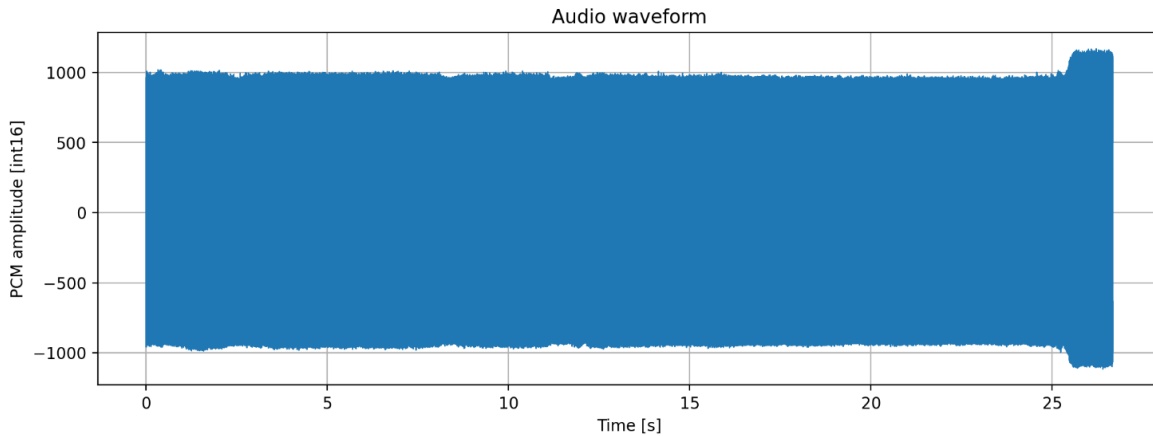


Figure 8, PCM amplitude with a tone of 1kHz acquired using the sensor board

Light Filter: Light data counts, see Figure 9, are normalized and then multiplied by their RGB value to obtain an RGB estimation of the observed light wave, see Figure 10.

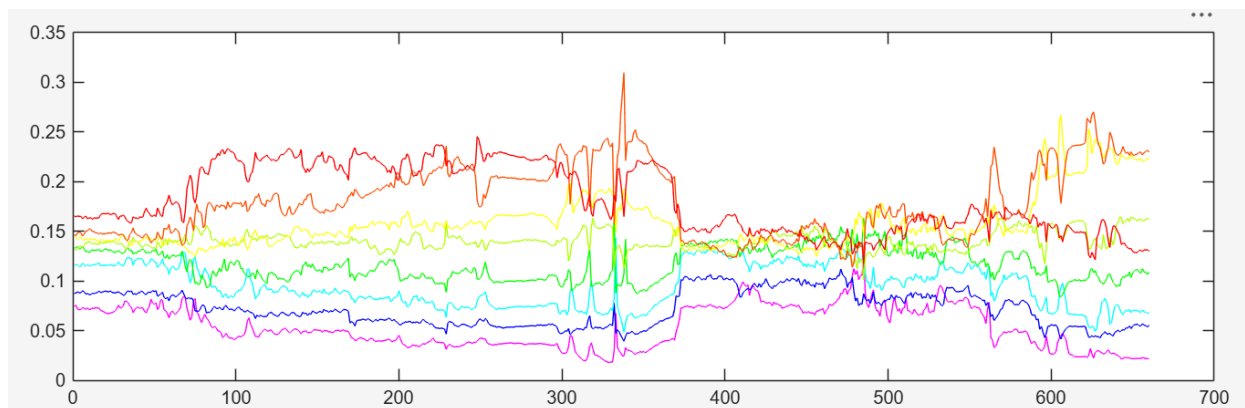


Figure 9, Raw counts of the 8 filters of the visible light

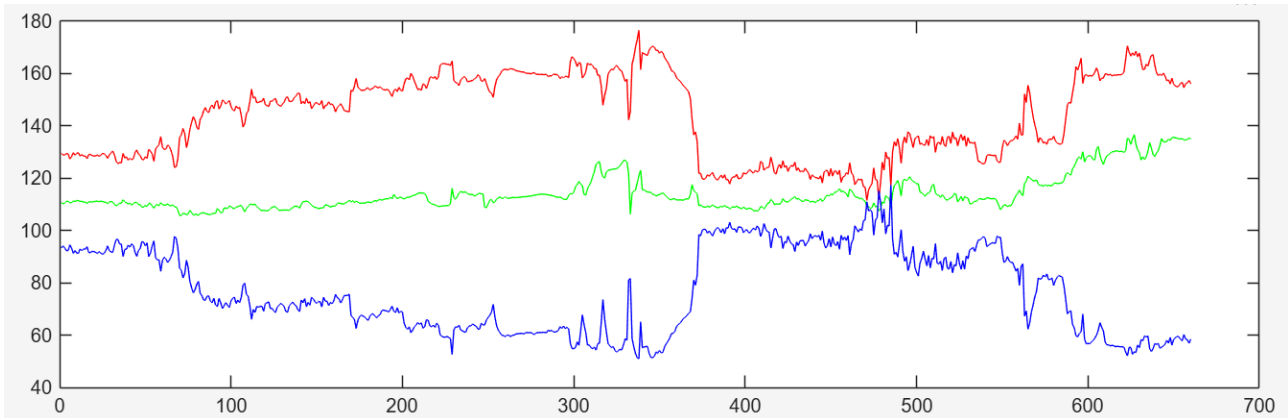


Figure 10, Correspondent RGB values

Then, the RGB value is saturated and transformed into their correspondent Temperature color value. See Figure 11.

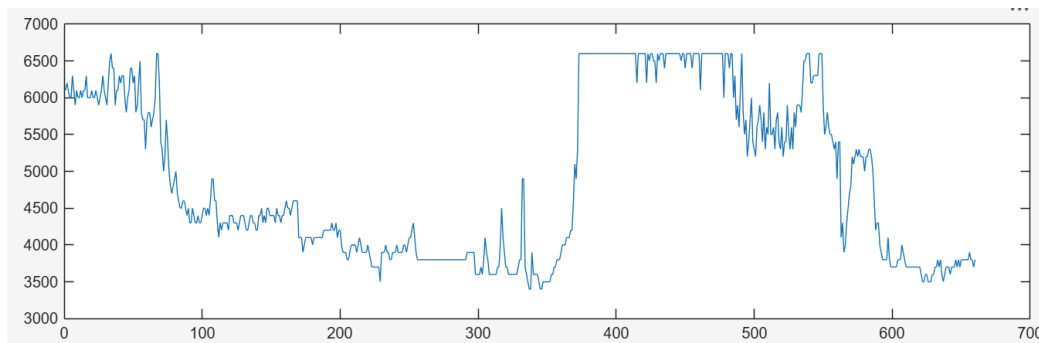


Figure 11, Light temperature values (in Kelvin)

Temperatures between 3500K - 5500K are classified as ideal for focus, 3000K - 3500K for creative tasks, and < 3000K for relaxation. High ranges (6000K - 6500K) and intense blue light exposure are monitored as triggers for short-term maximum focus that induce stress over time.^{4,5}

IMU and step counter: By filtering the raw counts from the gyroscope we can obtain an approximation of the current orientation of the glasses

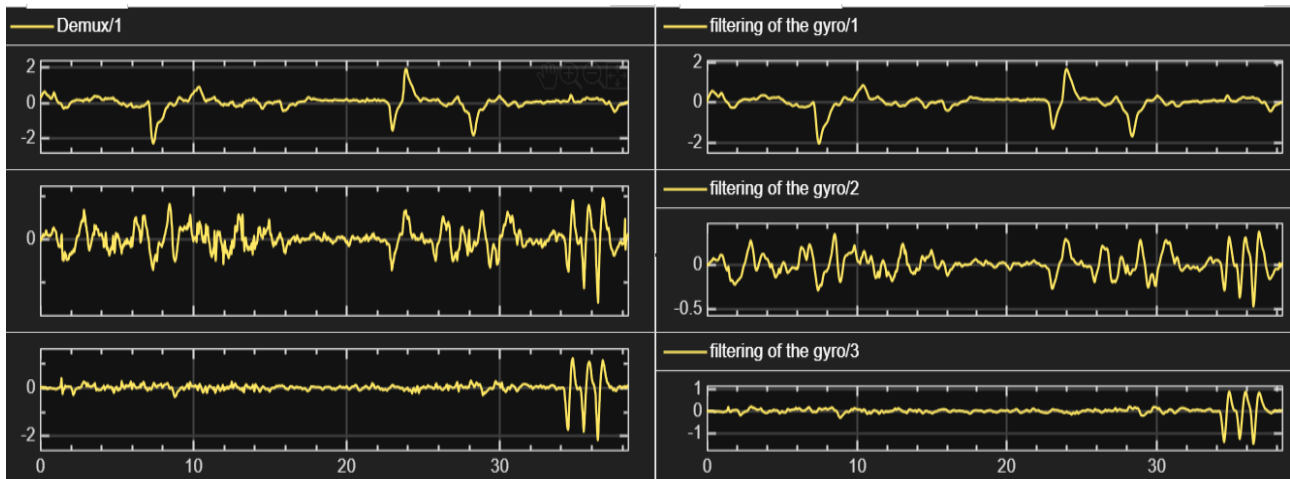


Figure 12, Raw gyro values vs filtered values (Bandwidth of 5 rad/s)

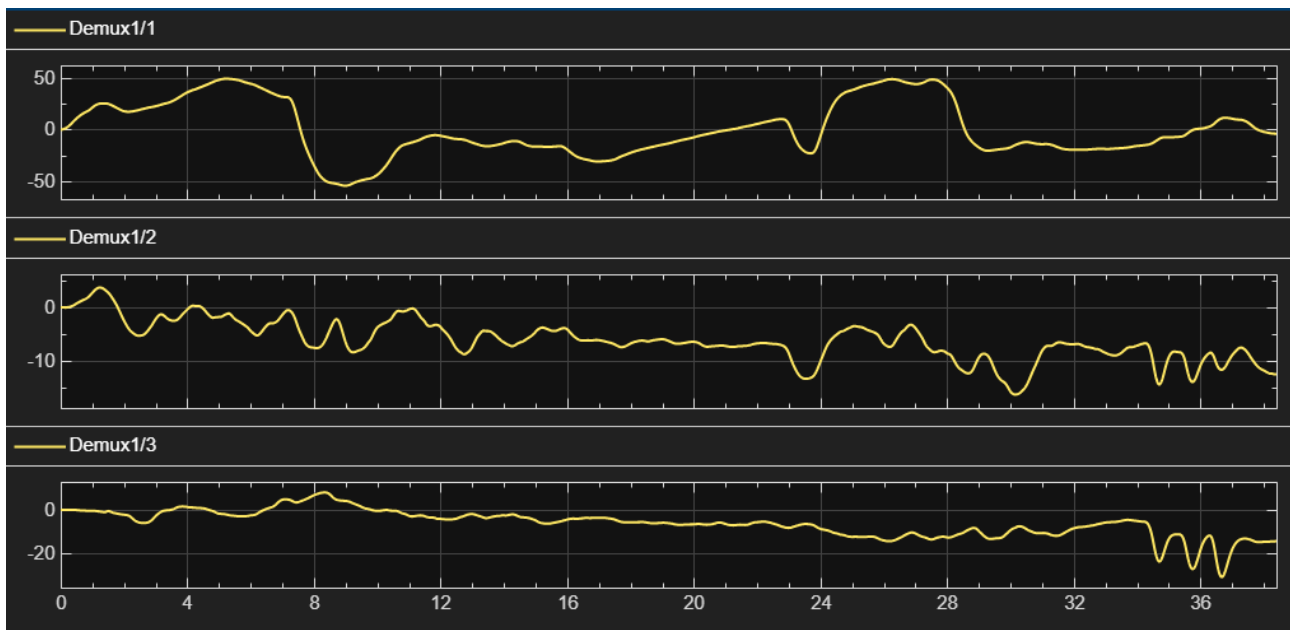


Figure 13, Angular displacement along the 3 axes of the glasses (in deg)

Then, with the orientation we can track the direction of the gravitational acceleration in real time and, in this way, remove its offset from the accelerometer data in x direction. See Figure 14, Original accelerometer data vs rescaled data (in m/s^2)

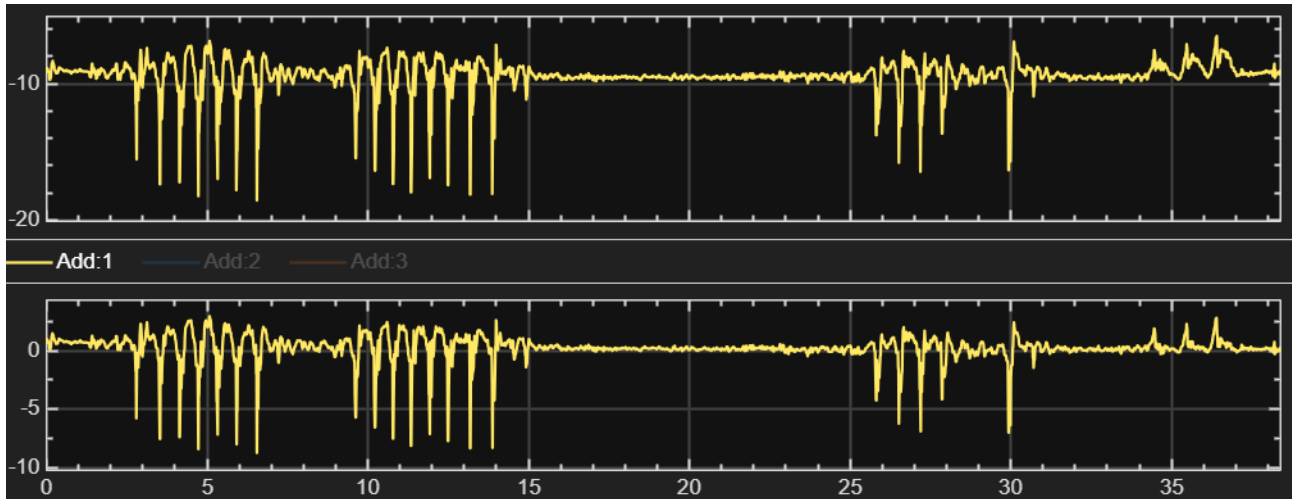


Figure 14, Original accelerometer data vs rescaled data (in m/s^2)

In the end, we can identify the steps by comparing the values of the rescaled `acc_x` data with a negative threshold. See Figure 15.

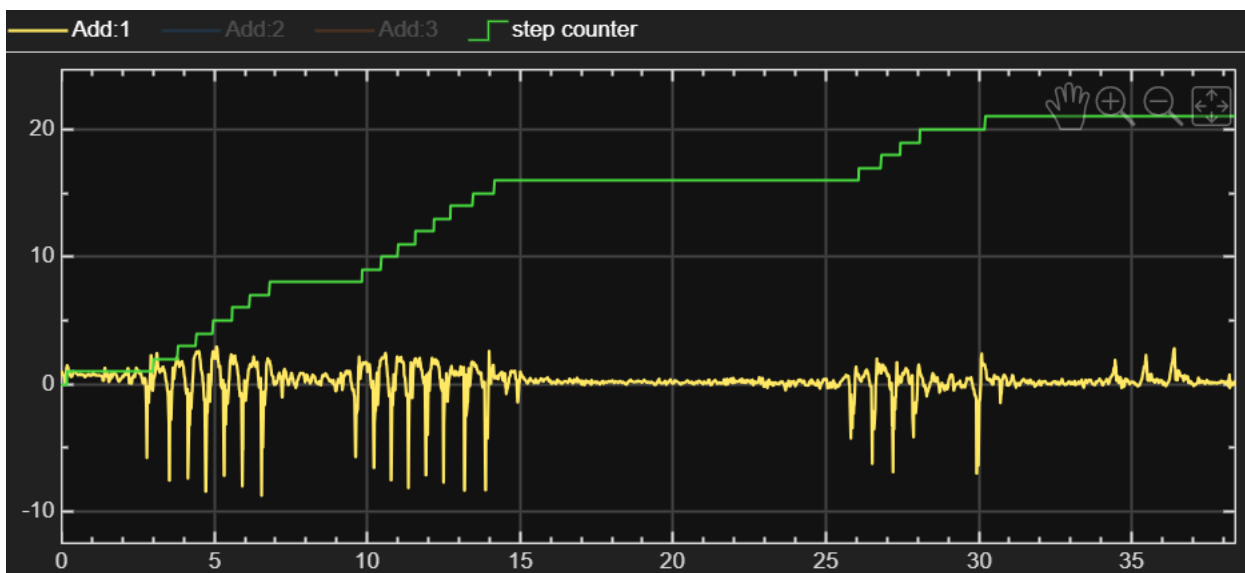


Figure 15, output of the step counter model after simulation

C Code Generation: the step counter and the color temperature identifier were both obtained as Simulink models; these can be easily translated to C code with the help of its extensions (Simulink coder, embedded coder, embedded coder support package) to implement the algorithm into the smart glasses' firmware. The generated files include a source file containing the algorithm, and several headers' files containing prototypes and data structures so that the firmware only needs to initialize the model, update its inputs, call the step function and read the output.

The embedded coder required some configurations, like setting imports and exports so they appear in the code and the target hardware, see Figure 16.

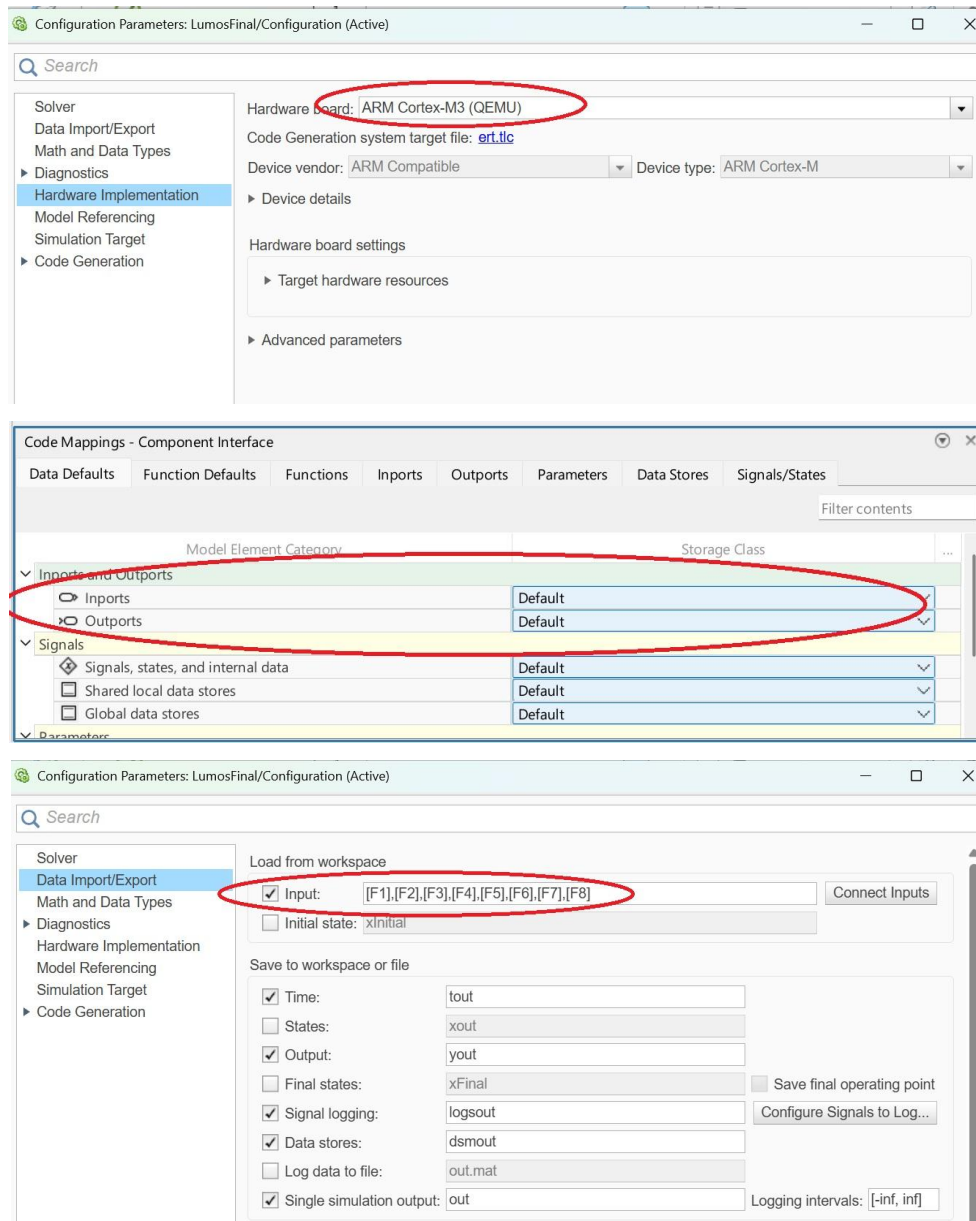


Figure 16, configuration settings for the Simulink embedded Coder

V. Optional Improvements - Improved Smartphone App

Lara Dau (20%), Luca Tufano (10%), Davide Venturini (20%), Youssef Gad (100%), Lorenzo Maccarini (0%)

We developed a new Flutter smartphone app to connect to smart glasses. We wanted the app to be simple and user-friendly for displaying all the different sensor data in real-time.

First, the app processes each unified 20-byte packet as we explained before in the mainboard firmware. This single packet contains all the data (steps, light, and audio). The app parses the data packet, according to the structure defined in the mainboard firmware, into each sensor metric and stores each to the corresponding variable in the SQLite database. We added the SQLite database directly into the app. This allows us to save the user's sessions locally on their phone so the user can look at their past fitness history and for future implementation to allow multiple users profile.

We also designed four main pages for the user interface. See Figure 17.

1. **Fitness Page:** Displays the step counts, the estimated the distance walked (by multiplying the stride length with the step count), and burned calories (using Mifflin-St Jeor equation) accordingly based on the users' entered data. Also, there is a weekly dashboard to show the step counts for each weekday.
2. **Light Page:** Displays the light sensor metrics as the Environment (Dark, Dim, Moderate, Bright, Very Bright), the Blue Light ratio to keep track of blue light exposure during the night. Also, to show the ambient color temperature in Kelvin.
3. **Stress & Noise Page:** Shows the microphone metrics like the environmental noise level (LAeq) in dB. And a session dose tracker to warn the user for prolonged exposure to loud sounds that could damage their hearing. Also, Concentration meter that helps the user decide the right place for studying as an example, showing the focus and stress indices.
4. **Settings Page:** Where the user will be able to input his/her details for better estimations of distance covered and calories burned. And change the theme of the app to Light/Dark.

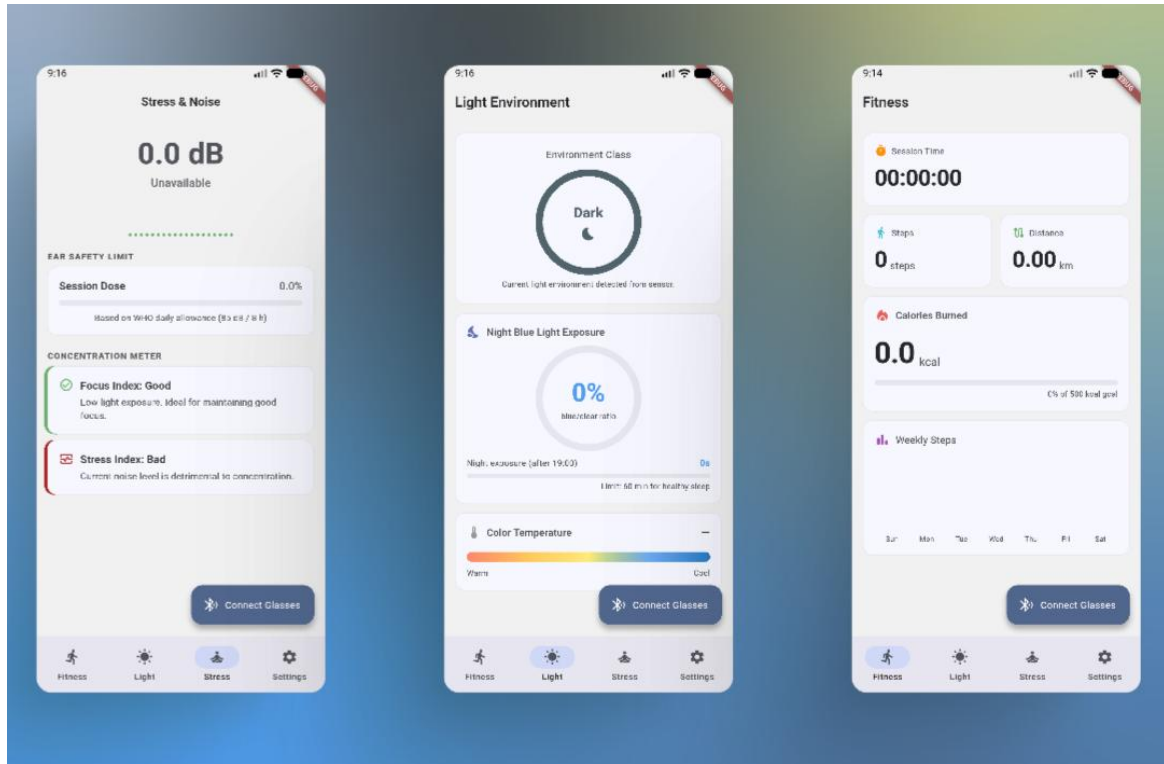


Figure 17, Screenshots of the Fitness Page, Light Page, and Stress Page

VI. Results and discussion

Lara Dau (50%), Luca Tufano (20%), Davide Venturini (60%), Youssef Gad (80%), Lorenzo Maccarini (90%)

The developed firmware converts raw sensor data into compact real-time features, enabling on-board monitoring, and live feedback through the companion application.

The results that we obtain can be divided into 3 steps:

First step: Each sensor was tested individually to verify that the acquisition chain was working correctly and that the produced data was consistent with the expected behavior.

Second step: After calibration of the microphone and light sensor, additional measurements were performed to validate the extracted features and refine the firmware parameters. For the light sensor, measurements were collected under different indoor illumination levels to verify the classification of the surrounding environment, see Figure 19. For the microphone, the calibrated A-weighted sound level was compared with the reference sound pressure level reported in the microphone datasheet at 93 dB SPL. The resulting relative error was approximately -4.45dB , see Figure 18. More data can be found in the attached "Validation DATA" folder.

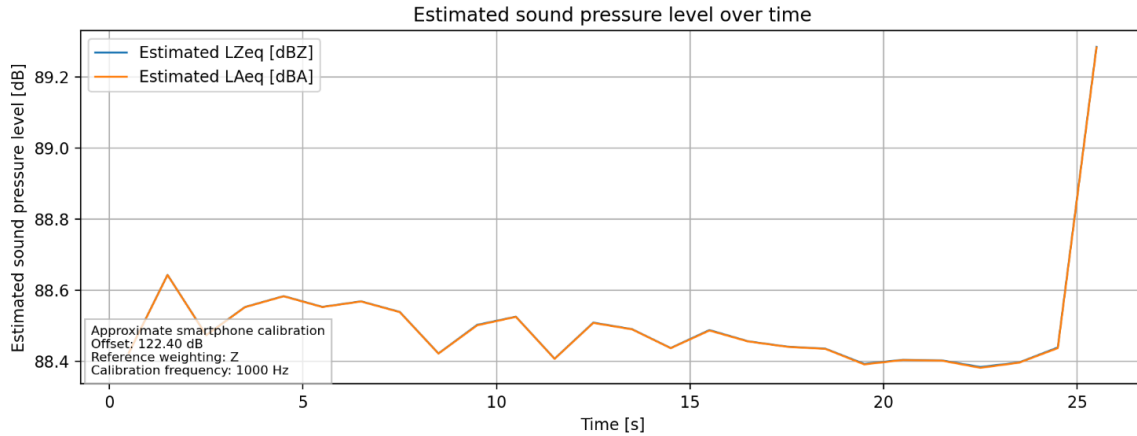


Figure 18, Estimation of the sound level dB-A

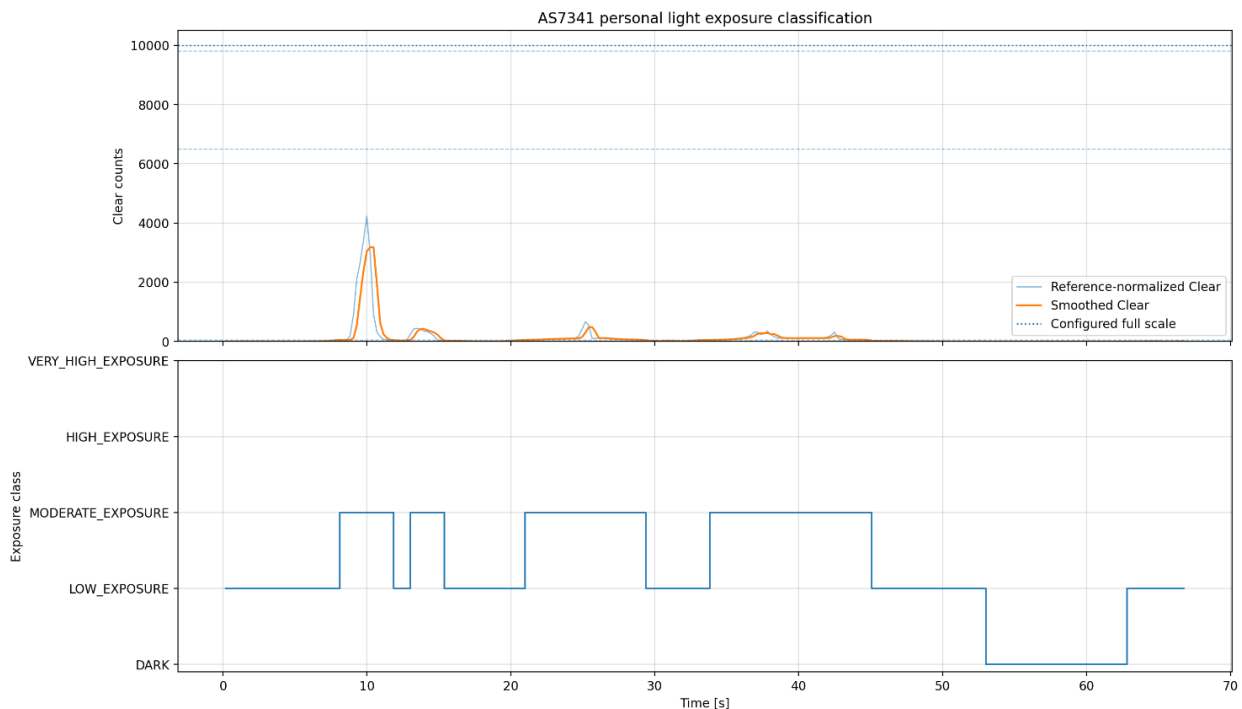


Figure 19, Light exposure based on the CLEAR channel, with different light illuminations settings

Third step: Our final custom app proves that smart glasses work in real-time. By keeping a steady Bluetooth connection with the glasses, the app tracks the environment in three main ways: the light around us, our physical activity (counting steps), and the noise levels. The app's dashboard easily shows all this information, giving a clear picture of our surroundings. In the end, these live results show that our app can successfully monitor the environment. By turning raw numbers from the sensors into simple advice, the app helps the user understand and change their surroundings. See Figure 20.

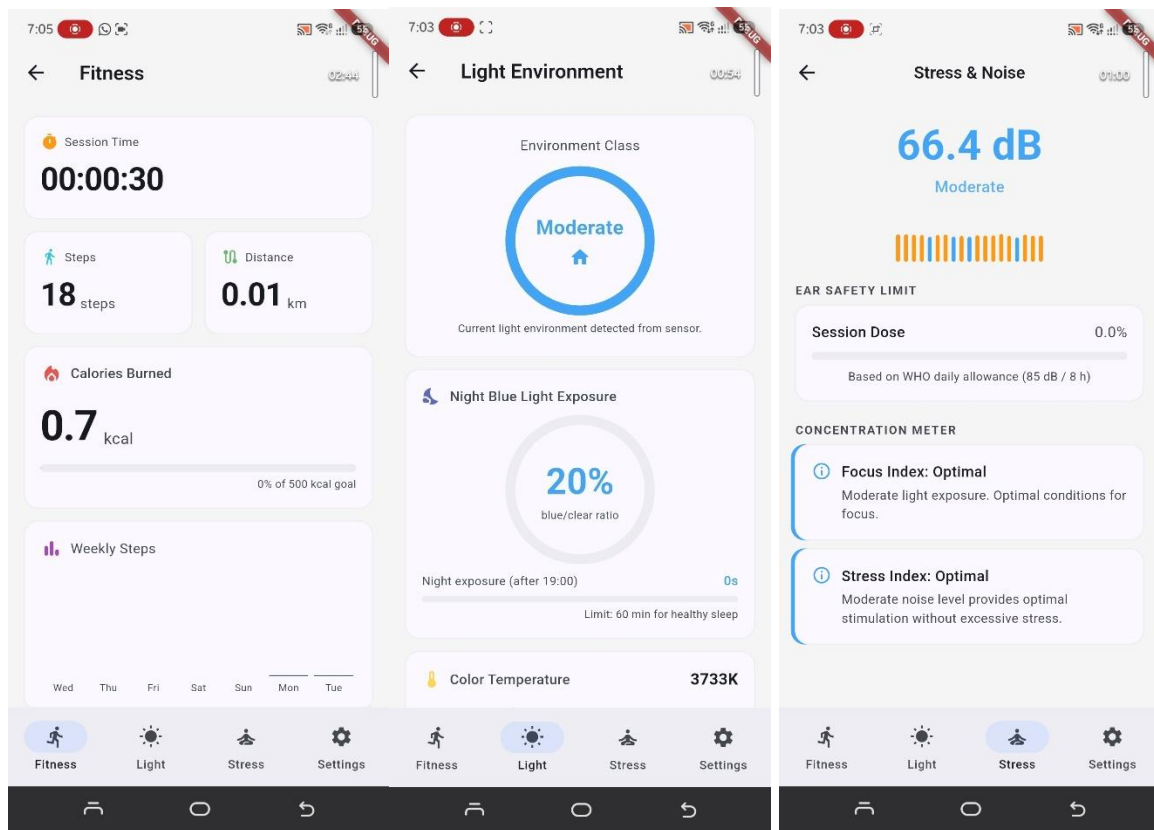


Figure 20, Live Screenshots from the app while testing.

Limitations and Future Improvements

The microphone was calibrated against a smartphone application rather than a professional sound level meter at a fixed 1 kHz frequency. Consequently, acoustic metrics are indicative and lack compensation for the microphone's frequency response above 10 kHz. Future work requires a frequency-dependent calibration curve. Similarly, the light sensor classification relies on empirical indoor observations rather than calibrated illuminance values. Utilizing a reference lux meter would allow thresholds to be defined in physical units, improving reliability. We plan to make the board use even less battery by taking advantage of the STM32's deep "STOP" modes. Right now, we only put the main processor into standard "SLEEP" mode. This is because the standard DMA (the hardware that moves the microphone audio into memory) needs the main system clocks to stay awake. In the future, we can utilize the Low Power DMA (LPDMA) to rewire the microphone and light sensor to use. So, we can keep collecting data in the background even when the main brain and clocks are completely shut off in deep STOP mode. Then, let a hardware timer or the IMU wake the CPU for just a millisecond to process the data and send the Bluetooth packet, which will

greatly save the battery life. Overall, the system provides compact and meaningful indicators for wearable monitoring, but further calibration and validation are essential for quantitative measurements.

Reference

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